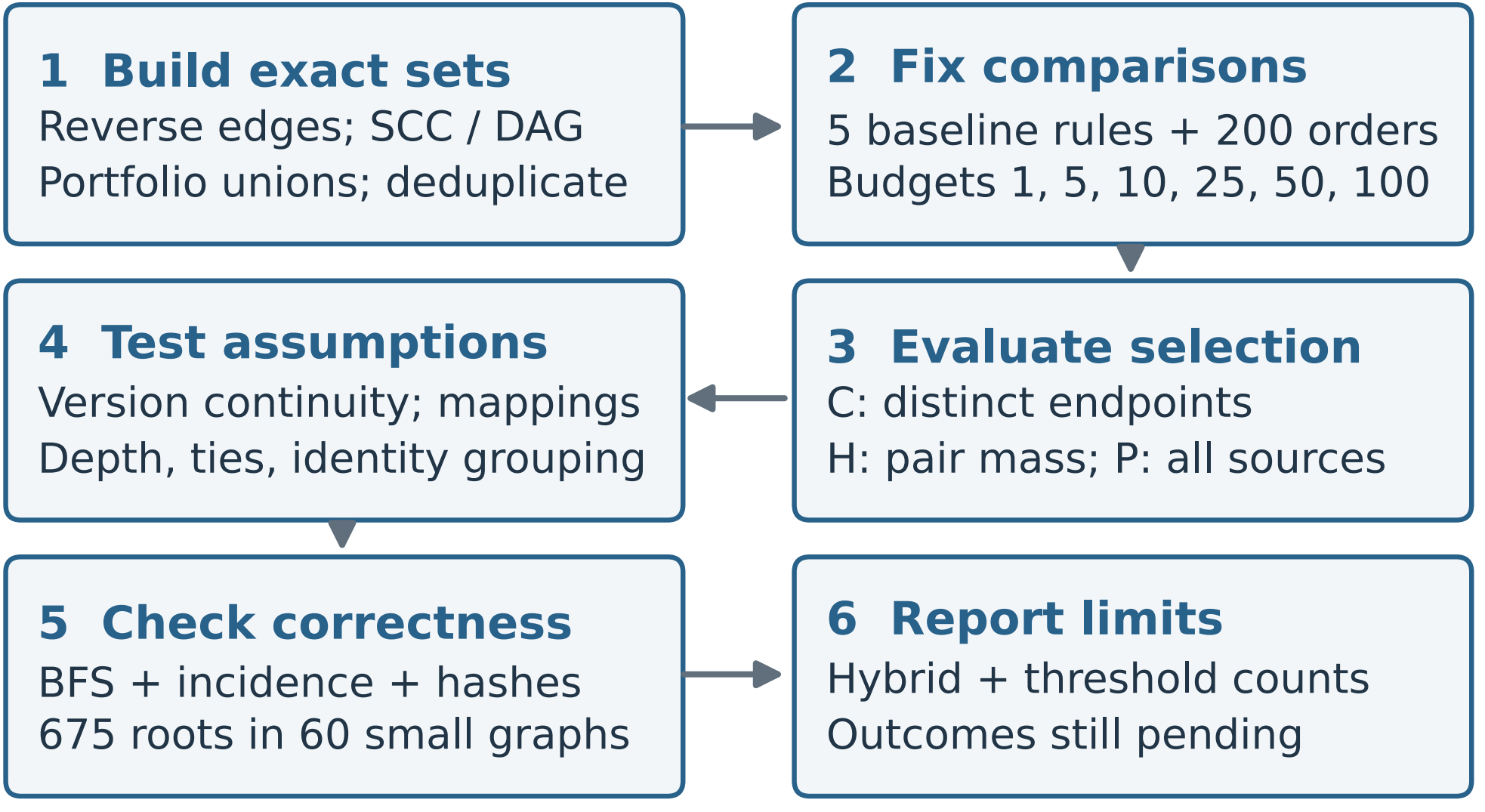ANALYSIS AND VALIDATION WORKFLOW